Nicholas Vietto

nicholasvietto@gmail.com | xxx-xxx-xxxx | Detroit, Michigan | Website | GitHub | LinkedIn

Technical Skills

- **Statistics**: Moderation/Interactions, Machine Learning, Predictive Modeling, Causal Inference, Experimental Design
- **Software**: Proficient in R, HTML/CSS, LaTeX, git, Bash/Zsh, Mplus; familiar with Python, SQL, Tableau, Julia, Typst
- **Tools**: Data Visualization, Package Development (R), Reproducible Workflows (Quarto), Dashboards, Data Wrangling

Professional Experience

University of Nebraska-Omaha

PhD Candidate — School of Criminology and Criminal Justice

August 2020 - Present

- Developed and compared 8 logistic regression and feedforward neural network models across 4 theoretical domains using {tidymodels}, predicting delinquency risk across 50+ variables in a ~2,000 participant sample, leveraging FFNNs to capture cross-domain interactions without explicit specification
- Compiled and merged multi-domain survey, psychological, and genetic data from the FFCWS across 5,000+ participants, subsampling and linking variables across domains for downstream modeling
- Designed, collected, analyzed, and published an original study (n=92) - managing IRB approval, recruitment, physiological and survey data collection, statistical analysis, and manuscript preparation - with all data and code publicly available
- Implemented the Johnson-Neyman technique in R to visualize interaction/moderation models, earning co-authorship on two peer-reviewed publications

University of Nebraska-Omaha

Graduate Teaching Assistant — School of Criminology and Criminal Justice

August 2020 - January 2026

- Designed and instructed 9 undergraduate courses (30+ students per section) centered on quantitative subjects, including statistics and research methods using Quarto and RMarkdown
- Developed and released two R packages to CRAN - {samplezoo} for streamlined dataset generation and {Rdune}, a ggplot2 extension providing custom color palettes for data visualization, both developed for and used in undergraduate instruction

Wayne State University

Data Analyst — Department of Criminology and Criminal Justice

August 2019 - August 2020

- Analyzed gene-environment interactions in R using regression modeling to predict behavioral outcomes
- Presented research on environmental adversity and genetic polymorphisms at a national academic conference

Education

University of Nebraska-Omaha

Ph.D. Criminology and Criminal Justice — Expected Spring 2027

Dissertation: Beyond Causal Explanations: A Data-Driven, Multidomain Approach to Predicting Risk for Delinquency

Wayne State University

M.S. Criminal Justice — 2020

Oakland University

B.A. Biological Science — 2016

Advanced Training

- Applied Bayesian Statistics for Social Scientists, ICPSR Summer Program at University of Michigan (2025)
- Future of Families and Child Well-Being Study Summer Data Workshop, Columbia University (2023)
- Machine Learning: Uncovering Hidden Structures in Data, ICPSR Summer Program at University of Michigan (2023)
- Longitudinal Structural Equation Modeling, Department of Psychology at University of Nebraska-Lincoln (2022)
- Structural Equation Modeling, Department of Psychology at University of Nebraska-Lincoln (2022)